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EDUCATION

Tecnológico de Monterrey

B.Sc. in Computer Science and Technology

Graduated in June 2025

95/100

WORK EXPERIENCE

Freelance Software Engineer

México

Delivered full-stack, ML, and automation projects for startups and organizations.

- Rlethoric (YC-backed startup): Integrated and unified internal systems for a LinkedIn marketing company. Led **full-stack** development, saving 1 hour per employee daily. (Aug 2025 - Ongoing)
- The Hub: Designed and deployed ML models improving food preparation quality by 10%. (Spring 2023)
- ArbidApps (Arbitrum Foundation initiative): Built a **no-code** platform enabling users to generate and deploy smart contracts on the Arbitrum blockchain. (Winter 2024)

Scholarvy

México

Co-Founder, CAIO, Lead Software Engineer

Summer 2023 - Oct 2024

- **Led a 10-member** development team building AI transparency tools for education, previously deployed in multiple countries across Latin America.
- Served as a **full-stack software engineer**, responsible for both front-end and back-end development, ensuring the delivery of a robust software solution.
- **Raised seed investment** with the government of Nuevo León and featured on an episode of **Shark Tank México**.

Research Fellowship, Tec de Monterrey

México

Research Engineer

Spring 2024 - Ongoing

- Developed a dual-autoencoder cryptographic system maintaining 98% reconstruction accuracy.
- Collected and analyzed EEG data for microsurgery applications using haptic robotic systems.
- Contributed to the design of **Deep Learning** models across multidisciplinary applications using PyTorch.

MAJOR PROJECTS

Scholarvy Ecosystem

May 2023 – Dic 2024

An AI in Education SaaS implemented across Latin America

- Architected Scholarvy's AI SaaS platform using Next.js, Node.js, and PostgreSQL on AWS; transitioned from multilayer to microservices architecture, improving scalability and uptime.

XAE²: Expansive Double Autoencoder

Aug 2024 – Oct 2024

Proposed a double autoencoder framework to optimize, compress and encrypt high-dimensional data

- Designed a dual-autoencoder framework for data compression and encryption, achieving 98% reconstruction accuracy on high-dimensional datasets.
- Demonstrated applications in secure data transmission and privacy-preserving ML pipelines across 1 experimental environment focused on secure communications.

Tren Maya's on-ride payment

Sep 2025 - Ongoing

Computational technologies and applied AI solutions in Industry.

- Engineered an offline POS with satellite link to enable in-ride payments in low-connectivity environments.

SKILLS

PROGRAMMING LANGUAGES

Advanced: TypeScript, JavaScript, Python
Intermediate: C++, Swift, Rust, Solidity, Matlab, C#, R

RELEVANT COURSEWORK (Tec de Monterrey)

Advanced Algorithm & Data Structures Design
Software Cloud Architecture
Multi-Agent Systems
Machine Learning

TECHNOLOGIES

Advanced: React, Next.js, Node.js, PostgreSQL, AWS, Git/GitHub
Intermediate: GCP, Azure, MongoDB, Express, Docker, PyTorch, OpenCV, Linux, Bash, Android SDK, Apple SDK, Unity3D, Three.js, Cypress, Jest

PROFILE

Full-stack and ML engineer with experience leading AI-driven startups, developing scalable SaaS solutions, and integrating applied AI systems across industry sectors.